

The Complete Sliding-Tile Solver Program

Let's begin by taking a look at the complete source code for the sliding-tile puzzle solver program. The rest of this chapter explains each section of code individually.

Copy the Python version of the code to a file named *slidingTileSolver.py*:

Python

```
import random, time

DIFFICULTY = 40 # How many random slides a puzzle starts with.
SIZE = 4 # The board is SIZE x SIZE spaces.
random.seed(1) # Select which puzzle to solve.

BLANK = 0
UP = 'up'
DOWN = 'down'
LEFT = 'left'
RIGHT = 'right'

def displayBoard(board):
    """Display the tiles stored in `board` on the screen."""
    for y in range(SIZE): # Iterate over each row.
        for x in range(SIZE): # Iterate over each column.
            if board[y * SIZE + x] == BLANK:
                print('__ ', end='') # Display blank tile.
            else:
                print(str(board[y * SIZE + x]).rjust(2) + ' ', end='')
        print() # Print a newline at the end of the row.

def getNewBoard():
    """Return a list that represents a new tile puzzle."""
    board = []
    for i in range(1, SIZE * SIZE):
        board.append(i)
    board.append(BLANK)
    return board
```